

Data Quality Report

Quality Score: 94%

Missing Values:

index 0 0 Company 0 1 Location_HQ 0 2 Industry 0 3 Laid_Off_Count 1253 4 Date 0 5 Source 0 6 Funds_Raised 390 7 Stage 0 8 Date_Added 0 9 Country 0 10 Percentage 1300 11 List_of_Employees_Laid_Off 0

Duplicates:

{'total_duplicates': np.int64(0)}

Outliers:

{'Laid_Off_Count': 279, 'Funds_Raised': 357, 'Percentage': 304}

AI Insight:

Based on the provided dataset quality metrics, here's a comprehensive analysis: #### 1. Data Quality Score - **Score:** 94% - **Analysis:** A score of 94% indicates that the dataset is of high quality. However, there is still room for improvement. The score suggests that most of the data is accurate, relevant, and consistent, but there are some areas that need attention, particularly regarding missing values. #### 2. Missing Values - **Missing Values Breakdown:** - **Laid_Off_Count:** 1253 missing values - **Funds_Raised:** 390 missing values - **Percentage:** 1300 missing values - All other fields have 0 missing values. - **Analysis:** The presence of missing values in key fields (Laid_Off_Count, Funds_Raised, and Percentage) can significantly impact data analysis. - **Laid_Off_Count:** Missing values here could skew any analysis aiming to understand workforce reductions. - **Funds_Raised:** Missing data in this field could affect financial analysis and investment insights. - **Percentage:** The missing values here may complicate any analysis that requires the percentage of layoffs relative to the total workforce or other metrics. #### 3. Duplicates - **Total Duplicates:** 0 - **Analysis:** The absence of duplicate records is a positive sign. It ensures that analyses and insights derived from the dataset are based on unique entries, which enhances the reliability of any conclusions drawn. #### 4. Outliers - **Outliers Identified:** - **Laid_Off_Count:** 279 outliers - **Funds_Raised:** 357 outliers - **Percentage:** 304 outliers - **Analysis:** Outliers can indicate anomalies or errors in the data, but they can also represent legitimate extreme cases. - **Laid_Off_Count:** A high count of outliers may reflect extreme layoffs in certain companies, which could skew average calculations. - **Funds_Raised:** Outliers in this category may represent companies that have raised significantly more capital than others, which could influence funding analysis. - **Percentage:** High numbers of outliers may suggest inconsistencies in how percentages are calculated or reported. #### Recommendations: 1. **Address Missing Values:** - Investigate the cause of missing values, especially for critical fields like Laid_Off_Count, Funds_Raised, and Percentage. - Consider imputation methods based on available data or consult with stakeholders to fill these gaps. 2. **Examine Outliers:** - Conduct a deeper analysis of the outliers to determine if they represent valid data points or if they are errors. - Depending on the findings, you may choose to exclude outliers from specific analyses or treat them differently. 3. **Continuous Monitoring:** - Establish a routine for assessing data quality, including monitoring for new missing values, duplicates, and outliers as the dataset grows or changes. 4. **Documentation and Data Governance:** - Maintain clear documentation on how data is collected, processed, and any transformations applied, to ensure

ongoing data quality and facilitate troubleshooting in the future. By addressing these areas, the overall quality and reliability of the dataset can be improved, leading to more accurate analyses and insights.